THE AGE POLYNOMIAL OF SQUAREFREE INTEGERS: LESLIE LINEARIZATIONS AND SPECTRAL LOCALIZATION OF MERTENS CANCELLATION

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ABSTRACT. For each cutoff x, the squarefree integers $n \le x$ are stratified by the additive age $\omega(n)$, the number of distinct prime factors. The age polynomial

$$F_x(z) = \sum_{k\ge 0} N_k(x) z^k = \sum_{n\le x} \mu^2(n) z^{\omega(n)} \tag{0.1}$$

contains the squarefree count at $z = 1$ and the Mertens function at $z = -1$. We develop a companion/Leslie linearization of F_x, an exact factorization of Mertens cancellation over its polynomial roots, and a numerical audit through $x = 10^7$. At 10^7 the closest root to -1 is $-0.9809394670636\ldots$; it accounts for 53.52% of the total logarithmic cancellation, while the three largest root contributions account for 84.78%. A local quadratic reconstruction using this root and the first two derivatives of F_x at -1 recovers $M(10^7) = 1037$ as $1036.512903\ldots$, a relative error of 4.697×10^{-4}.

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1. INTRODUCTION

Let μ denote the Möbius function and $\omega(n)$ the number of distinct prime divisors of n. Standard identities for μ, squarefree integers, and Dirichlet series are reviewed in [1, 2]. The starting point

Date: Version 2 — September 2026.

here is to retain the complete distribution of $\omega(n)$ over squarefree integers rather than collapsing immediately to the alternating sum $M(x)$.

The originating model is shown in Figure 1. Its demographic language motivates an age decomposition, while the mathematics below is formulated independently in terms of finite polynomials and companion matrices.

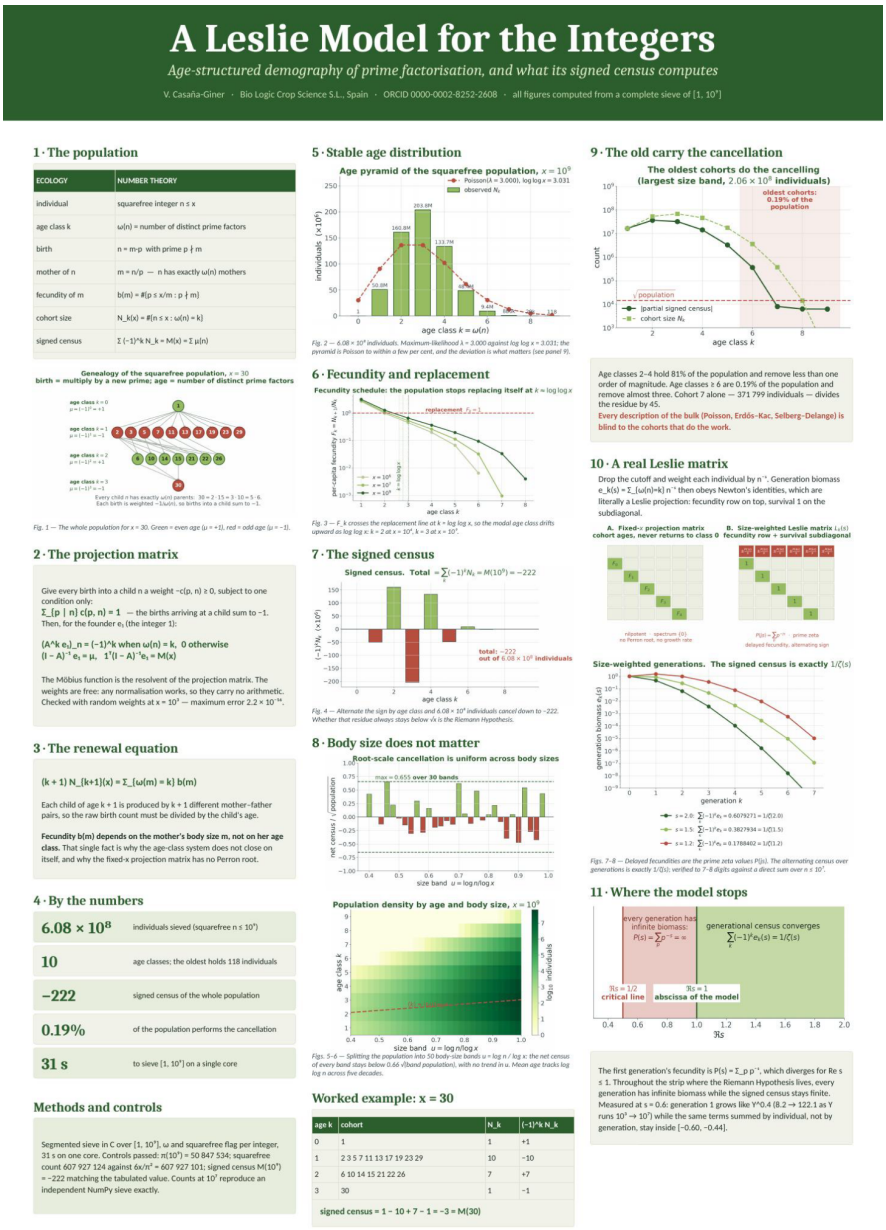


FIGURE 1. Source age-population model that motivated the polynomial and Leslie representations.

2. Age polynomial and exact endpoint identities

Definition 2.1. For $x \geq 1$ and $k \geq 0$, define the squarefree age cohort

$$N_k(x) = \#\{n \leq x : \mu^2(n) = 1,\ \omega(n) = k\}. \tag{2.1}$$

The associated age polynomial is

$$F_x(z) = \sum_{k=0}^{K(x)} N_k(x) z^k, \qquad K(x) = \max_{n \le x} \omega(n). \tag{2.2}$$

Proposition 2.2. *The two evaluations $z = 1$ and $z = -1$ give*

$$F_x(1) = Q(x) := \sum_{n \le x} \mu^2(n), \tag{2.3}$$

$$F_x(-1) = M(x) := \sum_{n \le x} \mu(n). \tag{2.4}$$

Proof. For squarefree n, one has $\mu(n) = (-1)^{\omega(n)}$, while nonsquarefree integers contribute zero to both μ and μ^2. Grouping by $\omega(n)$ gives (2.3) and (2.4). □

The degree $K(x)$ is the largest k for which the primorial $p_1 p_2 \cdots p_k \le x$. Thus the entire alternating sum over integers up to x is encoded by a polynomial of very low degree relative to x.

3. Root factorization and cancellation coordinates

Suppose

$$F_x(z) = N_K(x) \prod_{j=1}^{K} (z - r_j(x)), \qquad K = K(x). \tag{3.1}$$

Then the two endpoint values satisfy

$$Q(x) = N_K(x) \prod_{j=1}^{K} (1 - r_j), \tag{3.2}$$

$$M(x) = N_K(x) \prod_{j=1}^{K} (-1 - r_j). \tag{3.3}$$

Taking absolute values and dividing gives the exact identity

$$\frac{|M(x)|}{Q(x)} = \prod_{j=1}^{K} \frac{|1 + r_j(x)|}{|1 - r_j(x)|}. \tag{3.4}$$

Definition 3.1. For $M(x) \neq 0$, define the logarithmic contribution of a root by

$$c_j(x) = -\log \frac{|1 + r_j(x)|}{|1 - r_j(x)|}. \tag{3.5}$$

Then

$$\sum_{j=1}^{K} c_j(x) = -\log \frac{|M(x)|}{Q(x)}. \tag{3.6}$$

This is the precise meaning of the statement that a root controls a specified fraction of the cancellation: the fraction is measured in the additive logarithmic coordinate (3.6).

4. Leslie and companion linearizations

4.1. **The general degree-20 matrix.** Let

$$P(z) = a_{20}z^{20} + a_{19}z^{19} + \cdots + a_1 z + a_0, \qquad a_{20} \neq 0, \tag{4.1}$$

and set $b_j = a_j/a_{20}$. The following 20×20 Leslie-form companion matrix has unit transitions on the subdiagonal and signed first-row coefficients:

(4.2)

$$\mathcal{L}_{20}(P) = \begin{pmatrix}
-b_{19} & -b_{18} & -b_{17} & -b_{16} & -b_{15} & -b_{14} & -b_{13} & -b_{12} & -b_{11} & -b_{10} & -b_9 & -b_8 & -b_7 & -b_6 & -b_5 & -b_4 & -b_3 & -b_2 & -b_1 & -b_0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
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0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
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0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0
\end{pmatrix}.$$

Every displayed entry is explicit; the only implicit quantities are the normalized polynomial coefficients $b_j = a_j/a_{20}$. Direct companion-matrix algebra gives

$$\det(zI_{20} - \mathcal{L}_{20}(P)) = \frac{P(z)}{a_{20}}. \tag{4.3}$$

Thus the roots of P are precisely the eigenvalues of $\mathcal{L}_{20}(P)$.

4.2. **The exact matrix at ten million.** The computed cohorts at $x = 10^7$ are

$$(N_0, \ldots, N_8) = (1, 664579, 1903878, 2086746, 1103888, 286758, 32396, 1044, 1). \tag{4.4}$$

Hence

$$F_{10^7}(z) = z^8 + 1044z^7 + 32396z^6 + 286758z^5 + 1103888z^4 + 2086746z^3 + 1903878z^2 + 664579z + 1. \tag{4.5}$$

Its exact Leslie companion is

$$\mathcal{L}_8 = \begin{pmatrix}
-1044 & -32396 & -286758 & -1103888 & -2086746 & -1903878 & -664579 & -1 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0
\end{pmatrix}. \tag{4.6}$$

Consequently,

$$F_{10^7}(z) = \det(zI_8 - \mathcal{L}_8), \qquad M(10^7) = \det(-I_8 - \mathcal{L}_8) = 1037. \tag{4.7}$$

For a literal 20×20 realization of the same computed polynomial, set

$$\widehat{\mathcal{L}}_{20} = \mathcal{L}_8 \oplus 0_{12\times 12}. \tag{4.8}$$

Then

$$\det(-I_{20} - \widehat{\mathcal{L}}_{20}) = (-1)^{12} \det(-I_8 - \mathcal{L}_8) = 1037. \tag{4.9}$$

The zero block adds twelve zero eigenvalues and changes no endpoint magnitude or determinant here because 12 is even.

5. Dirichlet-series surface in age and size

For $\Re s > 1$, absolute convergence permits the exact Euler product

$$\mathcal{F}(s, z) = \sum_{n \geq 1} \frac{\mu^2(n) z^{\omega(n)}}{n^s} \tag{5.1}$$

$$= \prod_p (1 + z p^{-s}). \tag{5.2}$$

The signed section is

$$\mathcal{F}(s, -1) = \prod_p (1 - p^{-s}) = \frac{1}{\zeta(s)}. \tag{5.3}$$

Writing the prime zeta function as $P(s) = \sum_p p^{-s}$ in its half-plane of absolute convergence gives

$$\log \mathcal{F}(s, z) = \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} z^m P(ms), \tag{5.4}$$

whenever the double expansion is absolutely convergent. This supplies an exact two-variable age-size generating object; standard treatments of Euler products and multiplicative Dirichlet series may be found in [2].

6. Derivatives at the signed endpoint

Differentiating the finite polynomial gives the exact signed factorial moments

$$F_x^{(r)}(-1) = (-1)^r \sum_{n \leq x} \mu(n) (\omega(n))_r, \tag{6.1}$$

where $(u)_r = u(u-1)\cdots(u-r+1)$. In particular,

$$F_x'(-1) = -\sum_{n \leq x} \mu(n) \omega(n), \tag{6.2}$$

$$F_x''(-1) = \sum_{n \leq x} \mu(n) \omega(n) (\omega(n) - 1), \tag{6.3}$$

$$F_x'''(-1) = -\sum_{n \leq x} \mu(n) \omega(n) (\omega(n) - 1)(\omega(n) - 2). \tag{6.4}$$

For the present computation at $x = 10^7$,

$$F'(-1) = -51777, \quad F''(-1) = -273136, \quad F'''(-1) = -435972. \tag{6.5}$$

The Selberg–Delange method applied to (5.2) explains the natural scales obtained by differentiating the local parameter z near the negative integer -1; see [3]. The leading scales used in the numerical

audit are

$$F_x'(-1) \sim -\frac{x}{(\log x)^2}, \tag{6.6}$$

$$F_x''(-1) \sim -\frac{2x \log\log x}{(\log x)^2}, \tag{6.7}$$

$$F_x'''(-1) \sim -\frac{3x(\log\log x)^2}{(\log x)^2}. \tag{6.8}$$

At 10^7 the ratios of the measured derivatives to these leading scales are respectively

$$1.3451302545, \qquad 1.2762641757, \qquad 0.4885321447. \tag{6.9}$$

7. Critical root and local reconstruction of the Mertens function

Definition 7.1. A critical root $r_c(x)$ is a root of F_x minimizing $|1 + r|$. Write

$$h(x) = 1 + r_c(x). \tag{7.1}$$

Because $F_x(r_c) = 0$, finite Taylor expansion around -1 is exact:

$$0 = M(x) + \sum_{j=1}^{K(x)} \frac{F_x^{(j)}(-1)}{j!} h(x)^j. \tag{7.2}$$

Therefore

$$M(x) = -F_x'(-1)h(x) - \frac{1}{2}F_x''(-1)h(x)^2 - \sum_{j=3}^{K(x)} \frac{F_x^{(j)}(-1)}{j!} h(x)^j. \tag{7.3}$$

The linear and quadratic root estimates solve, respectively,

$$h_1(x) = -\frac{M(x)}{F_x'(-1)}, \tag{7.4}$$

$$M(x) + F_x'(-1)h_2(x) + \frac{1}{2}F_x''(-1)h_2(x)^2 = 0, \tag{7.5}$$

where the quadratic solution of smallest modulus is selected.

At 10^7,

$$r_c = -0.9809394670636025, \qquad h = 0.0190605329363975. \tag{7.6}$$

The linear prediction is

$$h_1 = 0.0200281978484655, \tag{7.7}$$

and the quadratic prediction is

$$h_2 = 0.0190690808540504, \tag{7.8}$$

with relative error

$$\frac{|h - h_2|}{|h|} = 4.484616291 \times 10^{-4}. \tag{7.9}$$

Using the actual h in the quadratic truncation of (7.3) gives

$$M_{(2)}(10^7) = 1036.512903023492, \tag{7.10}$$

so that

$$\frac{|M_{(2)}(10^7) - M(10^7)|}{|M(10^7)|} = 4.697174315 \times 10^{-4}. \tag{7.11}$$

8. Numerical experiment through ten million

Experiment 8.1. The object measured is the age polynomial (2.2), its roots, the root contributions (3.5), and the local reconstruction (7.3). The range is $100 \le x \le 10^7$ on 301 logarithmically spaced integer checkpoints. The expected outcome was to determine whether cancellation was distributed roughly evenly among the roots or concentrated in a small subset. The observed outcome is strong concentration in a few roots, together with a particularly stable real root close to -1 over most large checkpoints.

Table 1. Decade checkpoints. Percentages refer to cumulative logarithmic cancellation from the largest individual root contributions.

x	$Q(x)$	$M(x)$	degree	top 1 (%)	top 2 (%)	top 3 (%)
100	61	1	3	88.11	97.95	100.00
1 000	608	2	4	73.43	93.38	99.79
10 000	6 083	-23	5	42.31	84.62	97.11
100 000	60 794	-48	6	49.95	79.12	92.64
1 000 000	607 926	212	7	54.33	71.47	88.62
10 000 000	6 079 291	1037	8	53.52	69.15	84.78

At $x = 10^7$ the full root list, ordered by distance to -1, is shown in Table 2. The real root nearest -1 contributes 4.643706630653 units of the total logarithmic cancellation $8.676311\ldots$, namely 53.52%. The next two roots are a conjugate pair; the three individual roots together contribute 84.78%. Equivalently, grouping the conjugate pair as one real spectral mode, the first two modes account for 84.78%.

Table 2. Roots of F_{10^7} ordered by distance to -1. The last column is c_j from (3.5).

rank	$\Re r$	$\Im r$	$\lvert 1 + r\rvert$	c_j
1	-0.980939467	+0.000000000	0.0190605	4.643707
2	-1.590980522	+0.321309495	0.67268	1.356153
3	-1.590980522	-0.321309495	0.67268	1.356153
4	-0.000001505	+0.000000000	0.999998	0.000003
5	-2.662201276	+0.000000000	1.6622	0.789922
6	-4.731714296	+0.000000000	3.73171	0.429147
7	-20.167540266	+0.000000000	19.1675	0.099251
8	-1012.275642146	+0.000000000	1011.28	0.001976

Figure 2 follows the nearest-root distance, while Figure 3 records the fraction of logarithmic cancellation captured by one, two, and three roots.

At the sampled checkpoints, $M(x) = 0$ at

$$329,\ 355,\ 414,\ 541,\ 607,\ 825,\ 1525,\ 5208,\ 6072,\ 28184,\ 39811. \tag{8.1}$$

At each such checkpoint $F_x(-1) = 0$, so -1 is exactly a polynomial root.

Across all 301 checkpoints, the selected nearest root is real to numerical precision at 88.70% of the checkpoints. Restricting simultaneously to $M(x) \neq 0$ and to the derivative-based reconstruction mask gives 85.05% in the earlier reconstruction audit. The maximum observed absolute imaginary part of the selected nearest root is 0.4655711451. Over the last 100 checkpoints, the single-root share of total logarithmic cancellation ranges from 37.38% to 71.89%. These values show that the dominant-root phenomenon is persistent but not numerically constant.

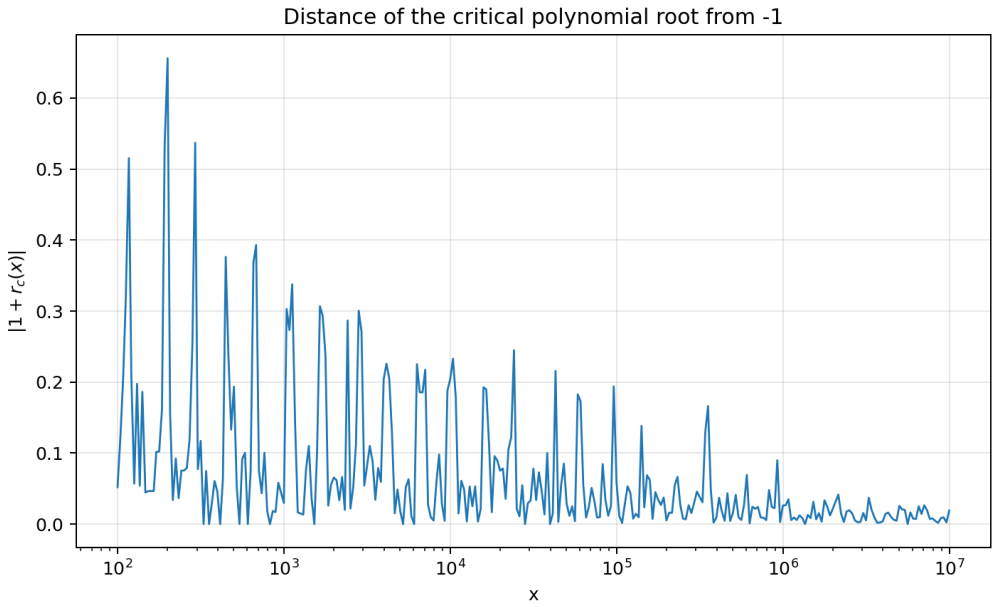


FIGURE 2. Distance $|1 + r_c(x)|$ at the 301 logarithmic checkpoints. Exact zeros of $M(x)$ correspond to a root at -1.

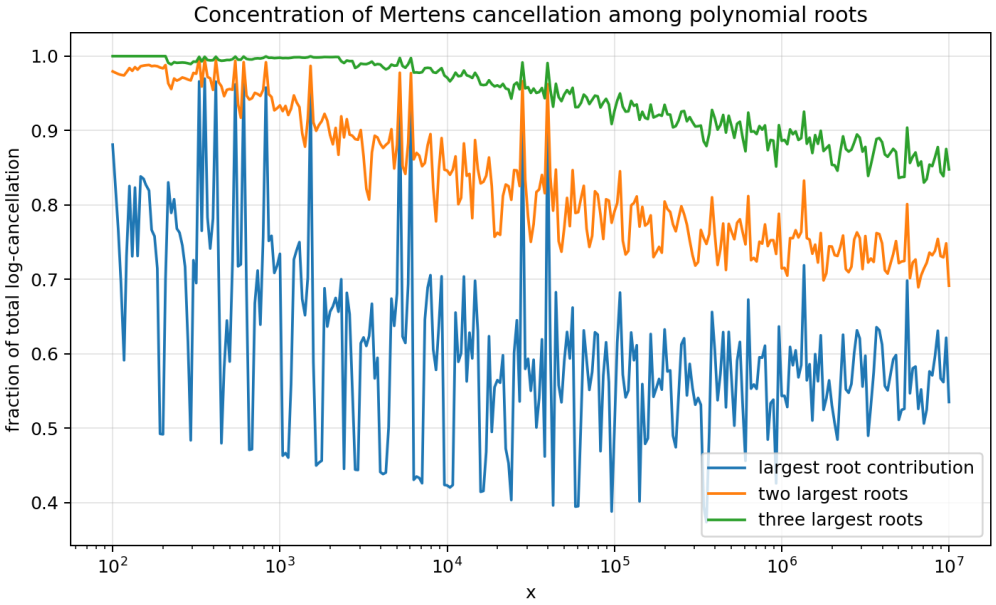


FIGURE 3. Cumulative fractions of total logarithmic cancellation accounted for by the largest individual root contributions.

9. Accuracy of the critical-root reconstruction

Table 3 summarizes the error distribution over checkpoints for which the selected critical root is real and $M(x) \neq 0$.

For $x \geq 10^6$, the median relative error of the quadratic root prediction is 1.3674×10^{-4}, and the median relative error in reconstructing $M(x)$ from the actual critical-root displacement and the

TABLE 3. Root and Mertens reconstruction error audit. The final two columns are relative errors for the quadratic reconstruction of $M(x)$.

$x \geq$	count	median linear h error	median quadratic h error	median quadratic M error	90% quadratic M error
10 000	168	4.892e-02	2.206e-04	2.210e-04	3.185e-03
100 000	120	4.333e-02	1.167e-04	1.247e-04	1.072e-03
1 000 000	61	3.560e-02	1.367e-04	1.327e-04	8.643e-04

first two derivatives is 1.3271×10^{-4}. The corresponding 90th percentile for the M reconstruction is 8.6432×10^{-4}.

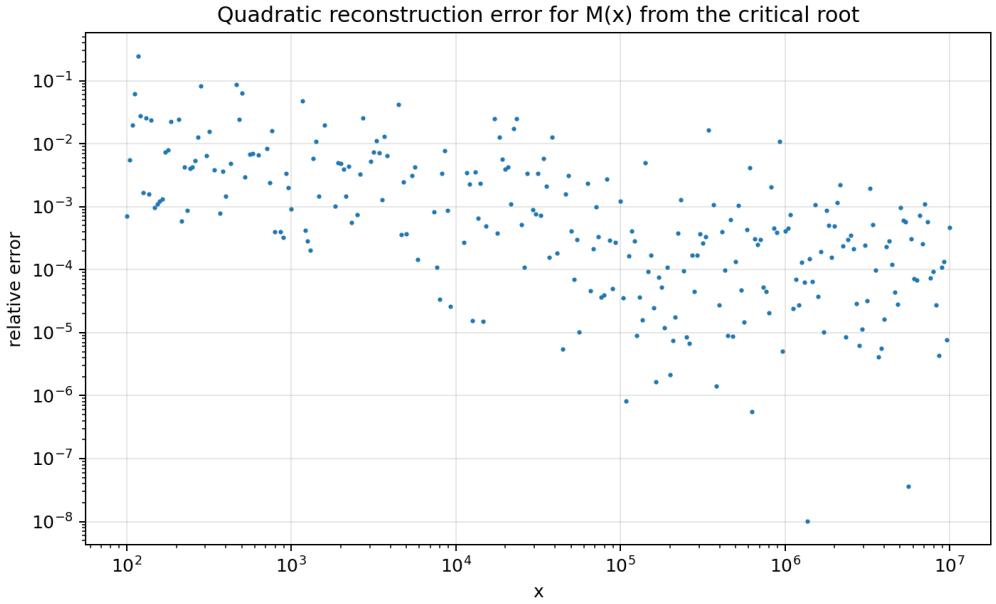


FIGURE 4. Relative error of the quadratic reconstruction of $M(x)$ at checkpoints with a real critical root and $M(x) \neq 0$.

The root cloud in Figure 5 shows that the root near -1 is geometrically separated from several much more negative real roots at many checkpoints, while complex conjugate pairs also occur.

10. A compact spectral coordinate for the Mertens fluctuation

The exact identity (7.3) separates the motion of the closest root from the signed factorial moments of the age distribution. If the higher Taylor terms are small on a range of x, the leading coordinate is

$$R_1(x) = -F'_x(-1)(1 + r_c(x)), \tag{10.1}$$

and the second-order coordinate is

$$R_2(x) = -F'_x(-1)(1 + r_c(x)) - \frac{1}{2}F''_x(-1)(1 + r_c(x))^2. \tag{10.2}$$

The numerical experiment shows that $R_2(x)$ tracks $M(x)$ to sub-per-mille relative accuracy over most of the largest sampled range. In this sense the irregular signed census is represented by a low-dimensional local spectral displacement together with two comparatively smooth signed age moments.

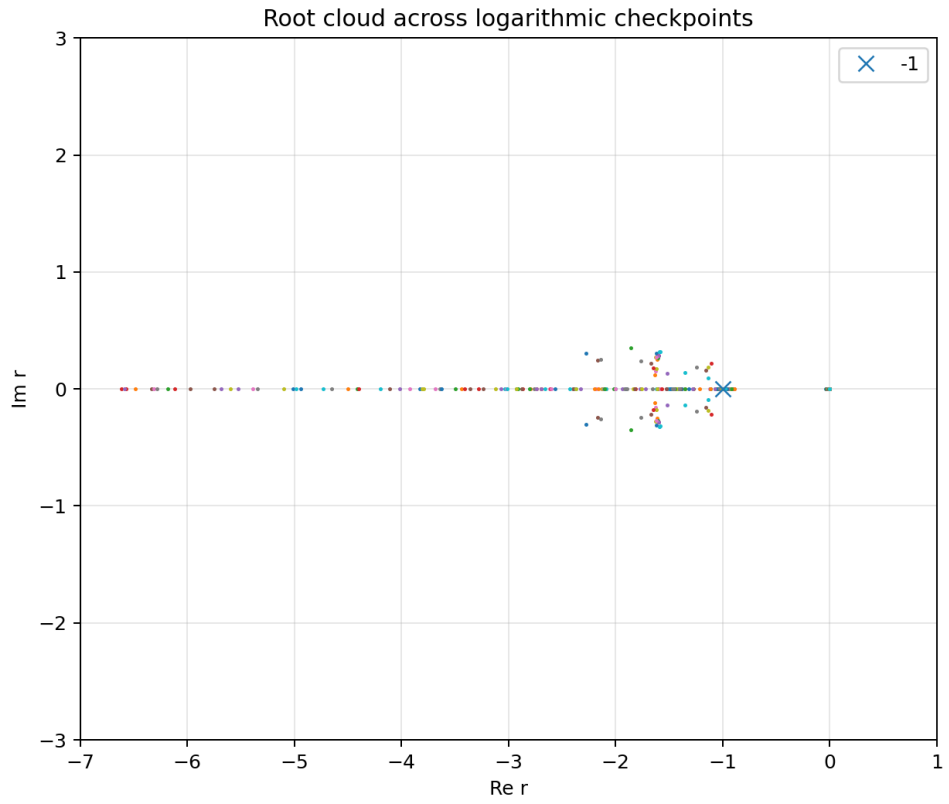


FIGURE 5. Polynomial roots over 70 representative logarithmic checkpoints. The plot is restricted to $-7 \le \Re r \le 1$ and $|\Im r| \le 3$ so that the local structure near -1 is visible; very large negative roots are omitted from the plotting window only.

Combining the leading scale (6.6) with (10.1) motivates the normalized displacement

$$\mathcal{R}(x) = \frac{x}{(\log x)^2}(1 + r_c(x)). \tag{10.3}$$

The numerical and asymptotic roles of (10.3) are distinct: the exact object is the root displacement $1 + r_c(x)$, while $x/(\log x)^2$ is the leading scale arising from the derivative at -1.

11. Conclusions

The age decomposition yields four concrete structures. First, $M(x)$ and the squarefree count $Q(x)$ are opposite endpoint evaluations of a single low-degree polynomial. Second, the roots of that polynomial are eigenvalues of an explicit Leslie-form companion matrix, giving the exact determinant identity $M(x) = N_K \det(-I - \mathcal{L}_K)$. Third, cancellation admits the additive root decomposition (3.6); at 10^7, one root contributes 53.52% and the first three roots 84.78% of the total logarithmic cancellation. Fourth, the nearest-root displacement around -1 admits the exact Taylor representation (7.3); at 10^7 its quadratic truncation reconstructs $M(x)$ with relative error 4.697×10^{-4}.

The next analytic object is therefore not merely the magnitude of $M(x)$ but the trajectory of the critical root and its interaction with the signed factorial moments $F_x^{(j)}(-1)$. The supplied data preserve the full cohort vectors and root information needed to extend the computation beyond 10^7 without changing the definitions used here.

References

[1] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, Oxford, 2008. DOI: 10.1093/oso/9780199219858.001.0001.

[2] H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory I: Classical Theory*, Cambridge Studies in Advanced Mathematics 97, Cambridge University Press, Cambridge, 2006. DOI: 10.1017/CBO9780511618314.

[3] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, 3rd ed., Graduate Studies in Mathematics 163, American Mathematical Society, Providence, RI, 2015. ISBN 978-0-8218-9854-3.

Appendix A. Python source: numerical reproduction

```
from pathlib import Path
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

ROOT = Path(__file__).resolve().parents[1]
DATA = ROOT / "data" / "leslie_integer_vectorized_1e7_augmented.npz"
OUT = ROOT / "outputs"
FIG = ROOT / "figures"
TAB = ROOT / "tables"
OUT.mkdir(exist_ok=True); FIG.mkdir(exist_ok=True); TAB.mkdir(exist_ok=True)

d = np.load(DATA, allow_pickle=False)
xs = d["checkpoints"].astype(np.int64)
counts = d["cohort_counts"].astype(np.int64)
roots_store = d["roots"]
Kgrid = np.arange(counts.shape[1])
Q = counts.sum(axis=1)
M = (counts * ((-1)**Kgrid)).sum(axis=1)
degree = np.array([np.max(np.flatnonzero(row)) for row in counts], dtype=int)

F1 = (counts[:,1:] * (Kgrid[1:] * ((-1)**(Kgrid[1:]-1)))).sum(axis=1)
F2 = (counts[:,2:] * (Kgrid[2:]*(Kgrid[2:]-1)*((-1)**(Kgrid[2:]-2)))).sum(axis=1)
F3 = (counts[:,3:] * (Kgrid[3:]*(Kgrid[3:]-1)*(Kgrid[3:]-2)*((-1)**(Kgrid[3:]-3)))).sum(axis=1)

critical = np.empty(len(xs), dtype=np.complex128)
delta = np.empty(len(xs))
root_rows = []
contrib_rows = []
shares = np.full((len(xs),3), np.nan)
mode1 = np.full(len(xs), np.nan)
mode2 = np.full(len(xs), np.nan)

for i,(x,k) in enumerate(zip(xs,degree)):
    coeff = counts[i,:k+1].astype(float)
    rr = np.roots(coeff[::-1])
    rr = rr[np.argsort(np.abs(rr+1))]
    critical[i] = rr[0]
    delta[i] = abs(1+rr[0])
    # cancellation decomposition: |M|/Q = product |1+r|/|1-r|
    if M[i] != 0:
        c = -np.log(np.abs(1+rr)/np.abs(1-rr))
        total = -np.log(abs(M[i])/Q[i])
        order = np.argsort(c)[::-1]
        cs = c[order]
        for j in range(min(3,len(cs))):
            shares[i,j] = cs[:j+1].sum()/total
    else:
        c = np.full(len(rr), np.inf)
    if x == 10_000_000:
        for j,z in enumerate(rr,1):
            root_rows.append((j,z.real,z.imag,abs(1+z),abs(1-z),(-np.log(abs(1+z)/abs(1-z)) if abs(1+z)>0 else
  np.inf)))
```

```
shift1 = -M/F1
shift2 = np.full(len(xs), np.nan)
for i in range(len(xs)):
    a = 0.5*F2[i]; b = F1[i]; c = M[i]
    if abs(a) < 1e-30:
        shift2[i] = -c/b if b else np.nan
    else:
        disc = b*b - 4*a*c
        if disc >= 0:
            hs = [(-b+np.sqrt(disc))/(2*a),(-b-np.sqrt(disc))/(2*a)]
            shift2[i] = min(hs,key=abs)

actual_h = critical.real + 1
realcrit = np.abs(critical.imag) < 1e-10
linear_relerr = np.full(len(xs), np.nan)
quad_relerr = np.full(len(xs), np.nan)
linear_relerr[realcrit] = np.abs(actual_h[realcrit]-shift1[realcrit])/np.maximum(np.abs(actual_h[realcrit]),1e
    -300)
quad_relerr[realcrit] = np.abs(actual_h[realcrit]-shift2[realcrit])/np.maximum(np.abs(actual_h[realcrit]),1e
    -300)
Mquad = -(F1*actual_h + 0.5*F2*actual_h**2)
Mquad_relerr = np.full(len(xs),np.nan)
mask = realcrit & (M!=0)
Mquad_relerr[mask] = np.abs(Mquad[mask]-M[mask])/np.abs(M[mask])

traj = pd.DataFrame({
    "x":xs,"Q":Q,"M":M,"degree":degree,
    "F1_minus1":F1,"F2_minus1":F2,"F3_minus1":F3,
    "rcrit_re":critical.real,"rcrit_im":critical.imag,"delta":delta,
    "root_linear_shift":shift1,"root_quadratic_shift":shift2,
    "linear_relerr":linear_relerr,"quadratic_relerr":quad_relerr,
    "M_quadratic_reconstruction":Mquad,"M_quadratic_relerr":Mquad_relerr,
})
traj.to_csv(OUT/"leslie_reproduced_trajectory_v2.csv",index=False)
pd.DataFrame(root_rows,columns=["rank_by_distance_to_minus1","root_re","root_im","abs_1_plus_root","
    abs_1_minus_root","log_cancellation_contribution"]).to_csv(OUT/"roots_x1e7.csv",index=False)

# Decade table with the stored cancellation shares from the original audit.
orig = pd.read_csv(OUT/"leslie_root_control_checkpoints.csv")
dec = orig[orig.x.isin([10**k for k in range(2,8)])].copy()
dec.to_csv(OUT/"decade_summary.csv",index=False)

# Error audit for x >= thresholds.
rows=[]
for th in [10_000,100_000,1_000_000]:
    m = mask & (xs>=th)
    rows.append({"threshold":th,"n_real_critical":int(m.sum()),
                 "median_linear_root_relerr":float(np.nanmedian(linear_relerr[m])),
                 "p90_linear_root_relerr":float(np.nanquantile(linear_relerr[m],.9)),
                 "median_quadratic_root_relerr":float(np.nanmedian(quad_relerr[m])),
                 "p90_quadratic_root_relerr":float(np.nanquantile(quad_relerr[m],.9)),
                 "median_M_quadratic_relerr":float(np.nanmedian(Mquad_relerr[m])),
                 "p90_M_quadratic_relerr":float(np.nanquantile(Mquad_relerr[m],.9))})
pd.DataFrame(rows).to_csv(OUT/"quadratic_error_audit.csv",index=False)
pd.DataFrame({"x":xs[M==0]}).to_csv(OUT/"zero_M_checkpoints.csv",index=False)

# Figures.
plt.figure(figsize=(8,5))
plt.semilogx(xs, delta, linewidth=1.1)
plt.xlabel("x")
plt.ylabel(r"$|1+r_c(x)|$")
plt.title("Distance of the critical polynomial root from -1")
plt.grid(True, alpha=.3)
plt.tight_layout(); plt.savefig(FIG/"critical_root_distance.png",dpi=180); plt.close()
```

```
plt.figure(figsize=(8,5))
plt.semilogx(orig["x"], orig["top1_root_frac"], label="largest root contribution")
plt.semilogx(orig["x"], orig["top2_roots_frac"], label="two largest roots")
plt.semilogx(orig["x"], orig["top3_roots_frac"], label="three largest roots")
plt.xlabel("x"); plt.ylabel("fraction of total log-cancellation")
plt.title("Concentration of Mertens cancellation among polynomial roots")
plt.legend(); plt.grid(True, alpha=.3); plt.tight_layout()
plt.savefig(FIG/"cancellation_concentration.png",dpi=180); plt.close()

plt.figure(figsize=(8,5))
mplot = mask & np.isfinite(Mquad_relerr)
plt.loglog(xs[mplot], Mquad_relerr[mplot], '.', markersize=3)
plt.xlabel("x"); plt.ylabel("relative error")
plt.title("Quadratic reconstruction error for M(x) from the critical root")
plt.grid(True, alpha=.3); plt.tight_layout()
plt.savefig(FIG/"quadratic_reconstruction_error.png",dpi=180); plt.close()

plt.figure(figsize=(7,6))
for idx in np.linspace(0,len(xs)-1,70,dtype=int):
    k=degree[idx]
    rr=np.roots(counts[idx,:k+1][::-1].astype(float))
    plt.plot(rr.real,rr.imag,'.',markersize=2)
plt.plot([-1],[0],'x',markersize=8,label='-1')
plt.xlim(-7,1); plt.ylim(-3,3)
plt.xlabel("Re r"); plt.ylabel("Im r")
plt.title("Root cloud across logarithmic checkpoints")
plt.legend(); plt.grid(True, alpha=.3); plt.tight_layout()
plt.savefig(FIG/"root_cloud.png",dpi=180); plt.close()

print(traj.iloc[-1].to_string())
print("Wrote outputs and figures.")
```

Appendix B. Python source: vectorized dataset construction

```
"""Build the integer-age data up to N using vectorized prime slices.
This is provided for full regeneration; the package also ships the computed NPZ.
"""
from pathlib import Path
import argparse, numpy as np

p=argparse.ArgumentParser(); p.add_argument('--N',type=int,default=10_000_000); p.add_argument('--out',default
    ='leslie_integer_vectorized_regenerated.npz'); a=p.parse_args()
N=a.N
isprime=np.ones(N+1,dtype=bool); isprime[:2]=False
for q in range(2,int(N**0.5)+1):
    if isprime[q]: isprime[q*q::q]=False
primes=np.flatnonzero(isprime)
omega=np.zeros(N+1,dtype=np.uint8)
squarefree=np.ones(N+1,dtype=bool); squarefree[0]=False
for q in primes:
    omega[q::q]+=1
    qq=int(q)*int(q)
    if qq<=N: squarefree[qq::qq]=False
# log-spaced checkpoints matching the audit
checkpoints=np.unique(np.rint(np.geomspace(100,N,301)).astype(np.int64)); checkpoints[-1]=N
maxk=int(omega[squarefree].max())
coh=np.zeros((len(checkpoints),maxk+1),dtype=np.int64)
for i,x in enumerate(checkpoints):
    vals=omega[1:x+1][squarefree[1:x+1]]
    coh[i]=np.bincount(vals,minlength=maxk+1)
np.savez_compressed(a.out,N=N,omega=omega,squarefree_packed=np.packbits(squarefree),checkpoints=checkpoints,
    cohort_counts=coh)
```

```
print(a.out)
```

Appendix C. Python source: core verification

```
from pathlib import Path
import numpy as np, pandas as pd
R=Path(__file__).resolve().parents[1]
d=np.load(R/'data'/'leslie_integer_vectorized_1e7_augmented.npz',allow_pickle=False)
c=d['cohort_counts'][-1].astype(int); k=np.arange(len(c)); Q=int(c.sum()); M=int((c*((-1)**k)).sum())
assert Q==6079291 and M==1037 and c.tolist()==[1,664579,1903878,2086746,1103888,286758,32396,1044,1]
r=d['critical_root'][-1]
assert abs(r.real+0.9809394670636025)<1e-12 and abs(r.imag)<1e-12
assert int(d['F1_minus1'][-1])==-51777 and int(d['F2_minus1'][-1])==-273136 and int(d['F3_minus1'][-1])
    ==-435972
print('Core x=10^7 audit: PASS')
print('Q=',Q,'M=',M,'critical root=',r)
```